# EXPERIMENT-1

## Linux Programming Environment & Basic Commands

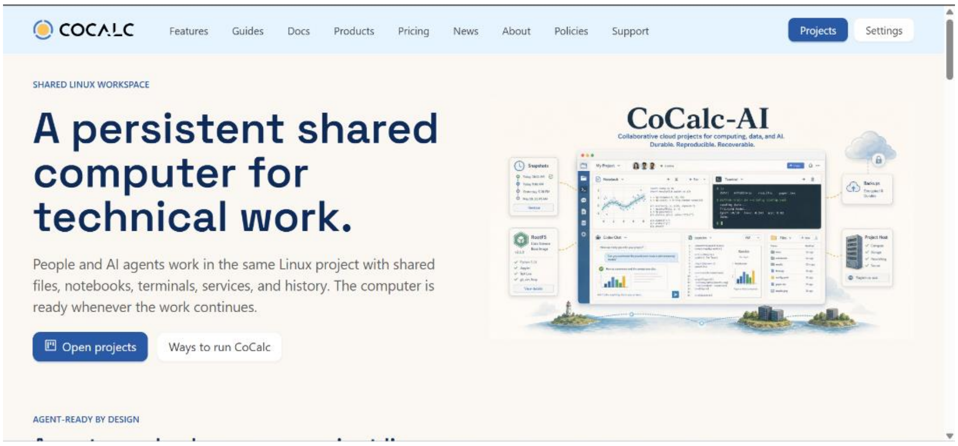
- **Setting up of Linux Programming Environment-** We can do C programming online or offline according to the needs. It is required for us to keep working on a Linux Environment.

1)-**Online**: Cocalc.ai

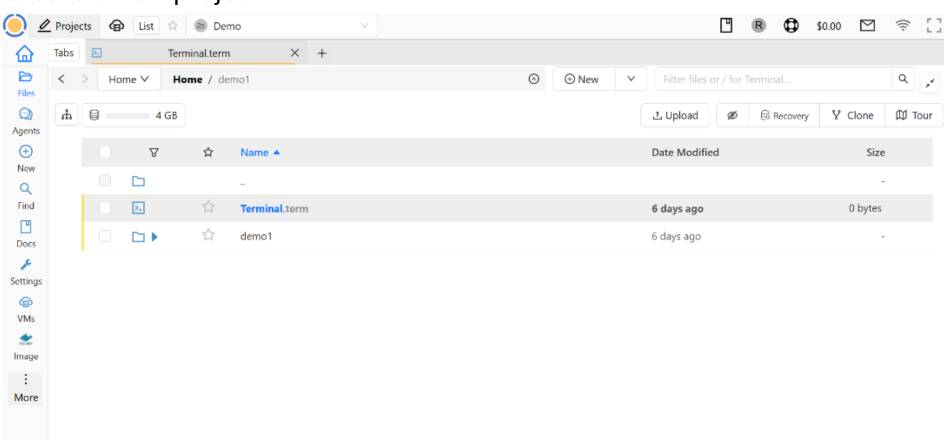
**CoCalc** is an online cloud-based platform where you can use a Linux terminal and programming environment through a web browser without installing Linux on your computer.

To use it we must follow the required steps-

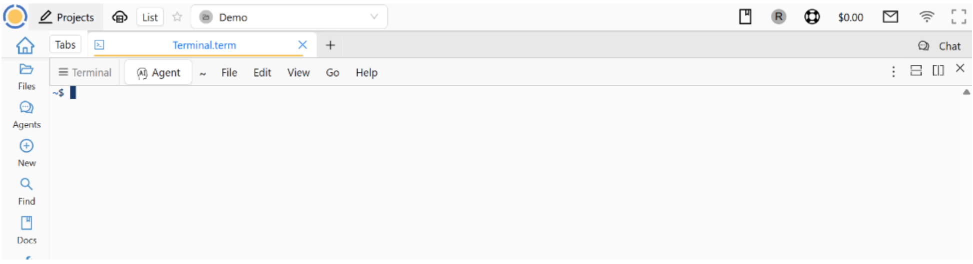
1. Open the CoCalc website in a web browser.



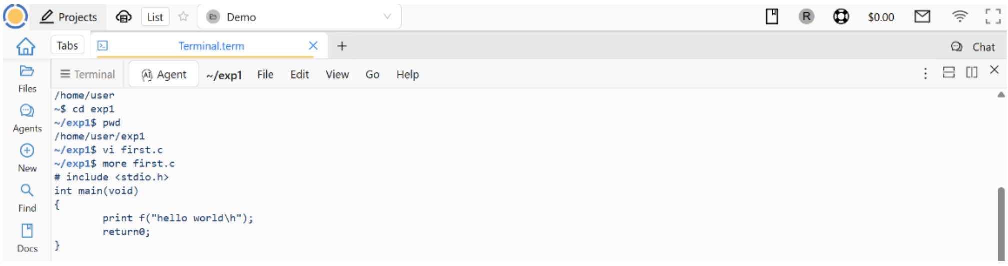

2. Create an account or sign in.
3. Create a new project.




4. Open a **Linux Terminal** inside the project.




5. The terminal provides a Linux environment.
6. Use commands such as pwd, ls, cd, mkdir, etc.




7. Create and compile C programs using CC.
8. Execute the compiled C program from the terminal.

```
~/exp1$ vi first.c
~/exp1$ cc first.c
~/exp1$ dir
a.out  first.c
~/exp1$ ./a.out
hello world
~/exp1$ █
```

**Advantage:** No Linux installation is required on the computer.

### 2. Offline methods of installing Linux

There are four methods mentioned by your professor.

**a) Dual Boot**

In a dual-boot system, **Windows and Linux are installed on the same computer**.

When the computer starts, you get a menu asking which operating system you want to use.

For example:

**Windows 11**

**Kali Linux**

You select whichever OS you want.

**Advantage:** Linux gets direct access to the computer's hardware and generally performs very well.

**Disadvantage:** Partitioning the disk incorrectly can cause data loss, so beginners need to be careful.

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**b) Single Boot**

In a single-boot system, **only one operating system is installed** on the computer.

For example, if you install Kali and remove Windows, the computer will directly start Kali.

**Advantage:** All system resources are available to Linux.

**Disadvantage:** You cannot directly use Windows unless you reinstall it or use virtualization.

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**c) Bootable Pendrive / Live USB**

A Linux ISO file is written to a USB pendrive, making it bootable.

You can then start your computer from the USB and run Linux.
There are two possibilities:

- **Live mode:** Linux runs from the USB without permanently installing it.
- **Installation:** Linux can be installed onto the computer from the USB.

**Advantage:** Useful for testing Linux or installing it.
**Disadvantage:** Running Linux directly from a USB is generally slower than running it from an internal SSD.

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**d) Virtual Machine (VM)**
A **Virtual Machine** allows you to run another operating system inside your existing operating system.
For example:
**Windows → Oracle VirtualBox → Kali Linux**
Your Windows system continues running normally while Ubuntu runs inside a virtual machine.
This is the method I recommend for your experiment because **you don't need to modify your Windows installation or partition your SSD**.
**3. Virtual Machine Manager / Hypervisor**
A **hypervisor** is software or firmware that creates and manages virtual machines.
It allows one physical computer to run multiple operating systems virtually.
There are two major types.
**Type-1 Hypervisor — Bare Metal**
A Type-1 hypervisor runs **directly on the physical hardware**, without requiring a conventional host operating system underneath it.
**Examples:**

- VMware ESXi
- Microsoft Hyper-V Server
- Xen

**Simple diagram:**
**Hardware**
↓
**Type-1 Hypervisor**
↓
**Virtual Machines**
**Where is it used?**
Type-1 hypervisors are commonly used in **servers, data centres and enterprise environments** because of their performance and efficient resource management.

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**Type-2 Hypervisor — Hosted**
A Type-2 hypervisor runs **on top of an existing operating system**.
In your case:
**Hardware**
↓
**Windows**
↓
**Oracle VirtualBox**

↓

**Kali Linux VM**

**Examples:**

- Oracle VirtualBox
- VMware Workstation
- VMware Fusion

Your professor has specified that you will use **Oracle VirtualBox**.

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### 4. Installation of VMM and VM

Now comes the actual practical work.

You need two things:

**Software 1 — Oracle VirtualBox**

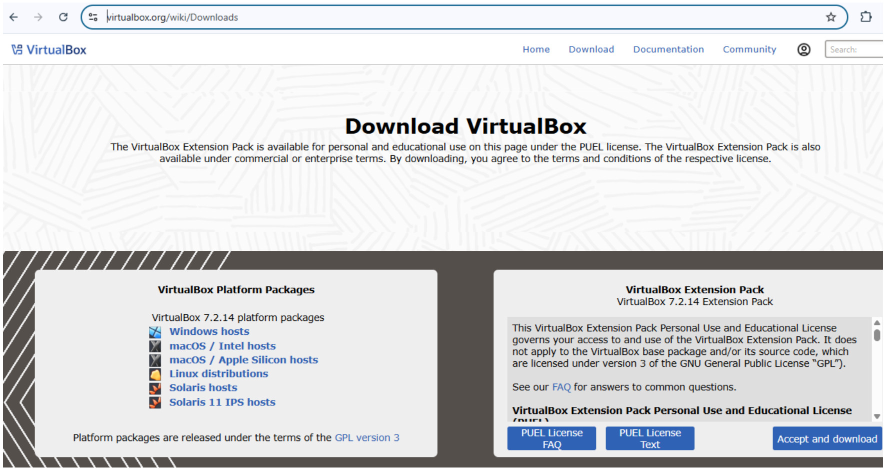
This is your **Type-2 hypervisor**.

**Software 2 —Kali Linux ISO**

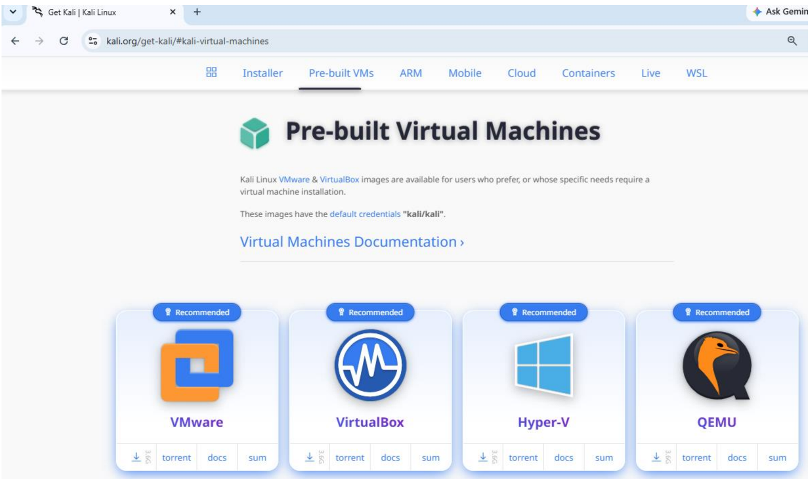
This is the operating-system installation file that you will use to create your Linux VM.

- # <u>Installation of VMM and VM</u>

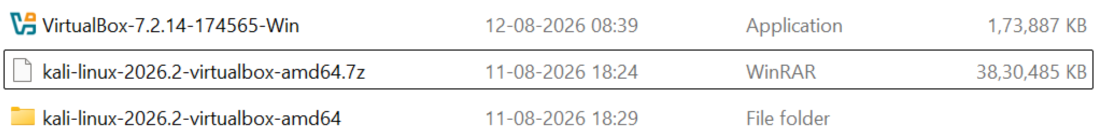
- Step-1: Download Oracle Virtual Box for “Windows hosts”



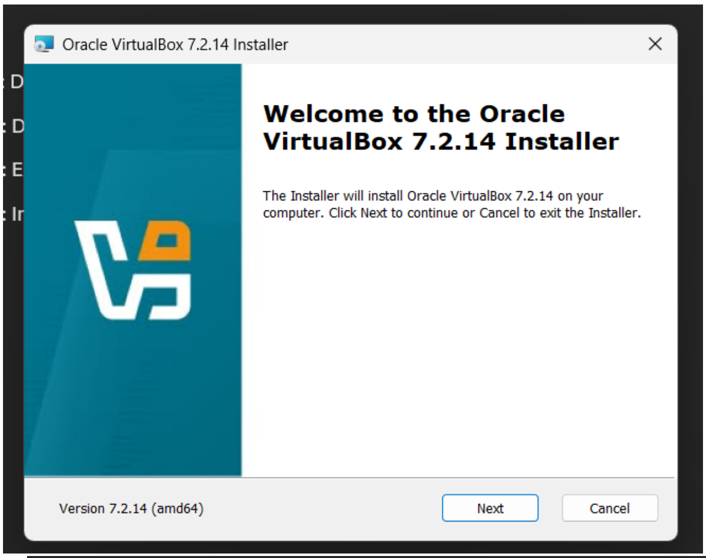

- Step-2: Download Pre-built Kali Linux VM for VirtualBox

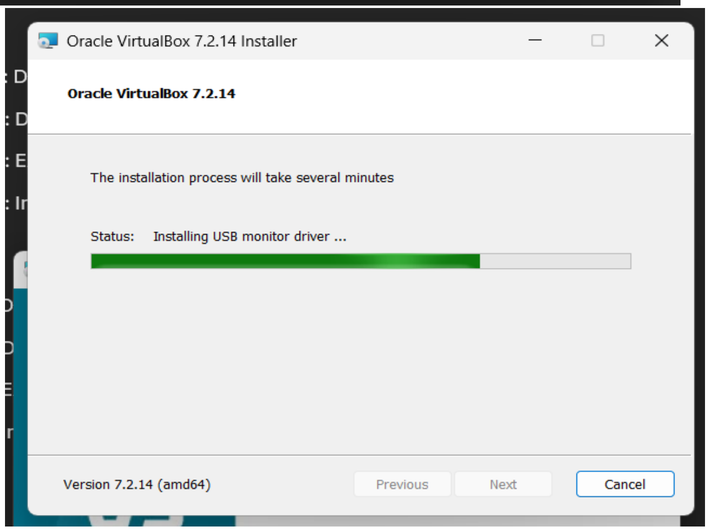



- Step-3: Extracted the Kali Linux downloaded file

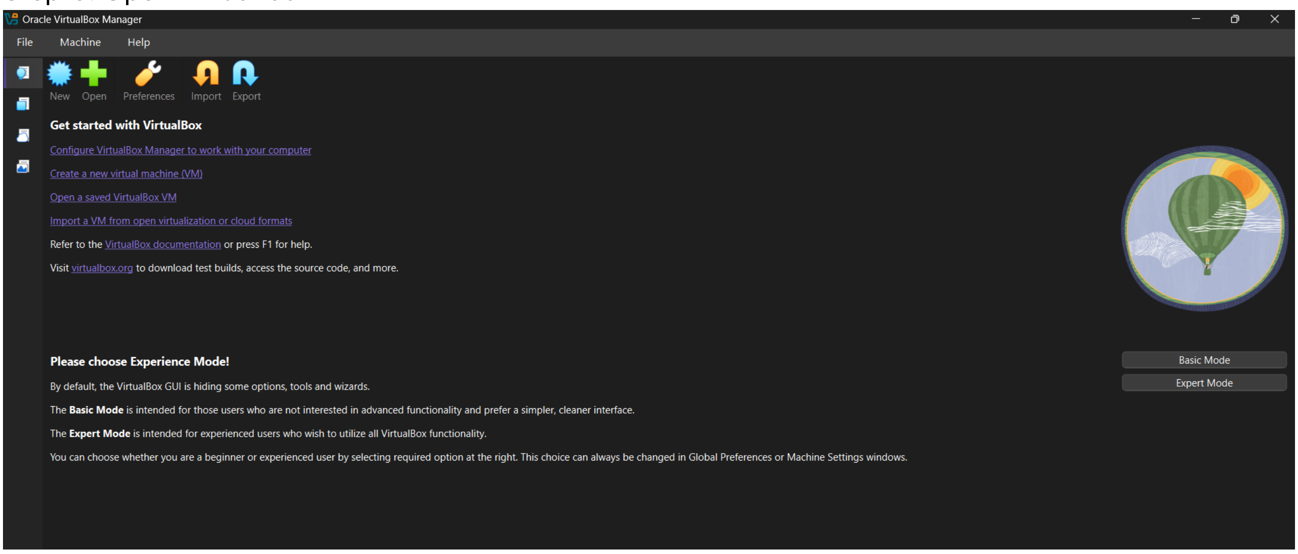



- Step-4: Install VirtualBox

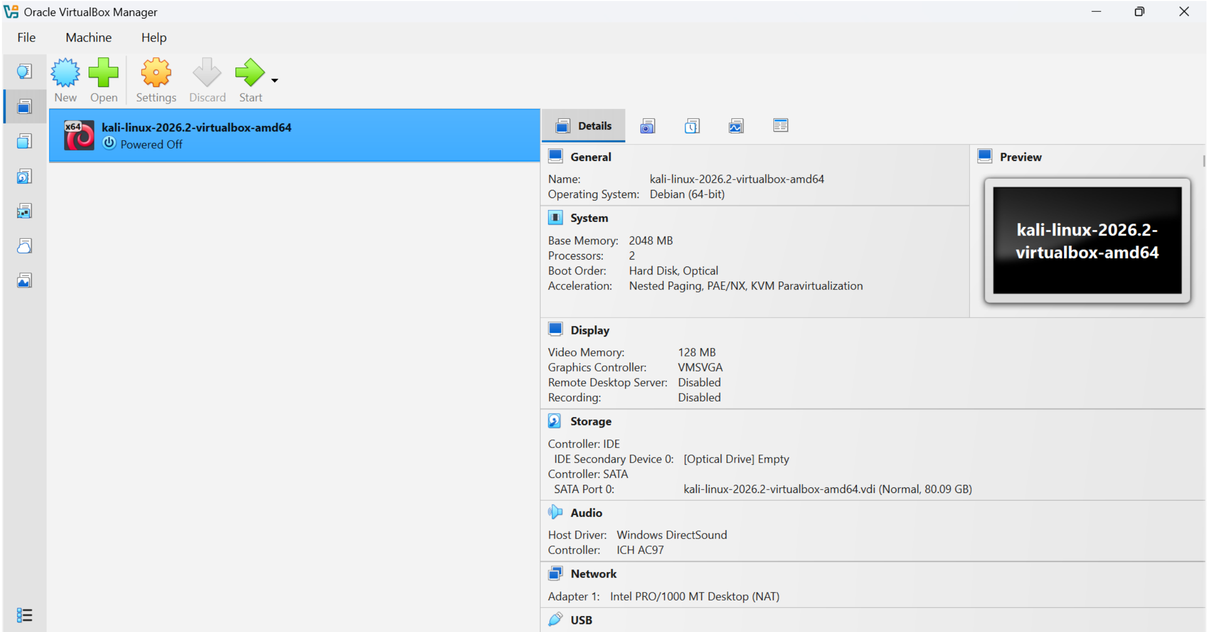


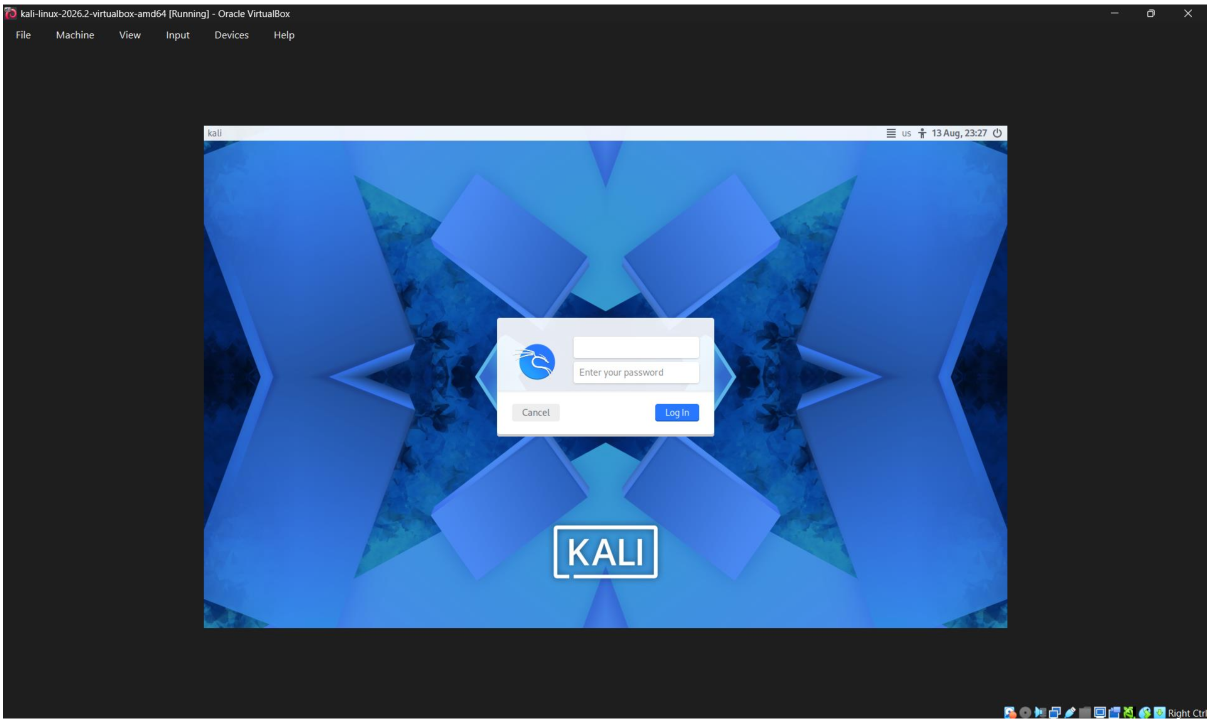

- Step-5: Open Virtual box



- Step-6: Open the “saved VM”




- Step-7: Start the VM




- Thus, the installation of VMM (Hypervisor Type-2 Oracle VirtualBox) & VM (Kali Linux) is completed.
- Step-8: Open terminal and create the following simple program, compile and execute to understand the working of the environment.

## B) <u>Basic Linux Commands</u>

- **<u>Terminal basics</u>**
  1) Pwd- shows the current directory you are working on
  2) Mkdir- creates new directories
  3) Rmdir-removes the empty directory you want
  4) Ls-lists all the file and folder of your directory
  5) Cd-changes directories
  6) Clear clears the full terminal you are working on
  7) History-to view all the recent inputs
  8) Man-it displays help required for your commands
  9) Whoami- shows the current username of the user. e.g. in kali linux it shows kali
- **<u>Files</u>**
  1) Vi- This command creates a C file
  2) File- It shows the extension of the file
  3) Rm- remove files
  4) Cp-copies files
  5) Mv- move files
  6) Cat- Displays file contents
  7) More- to see your commands in your file
  8) Less- used to view contents
  9) Head-outputs the first part of file
  10) Tail-outputs the last part of file

  - **<u>>Output redirection:</u>**

Sends the program's output to a file. If the file already exists, it overwrites it.

./program > output.txt

<u>>> **Append redirection:**</u>

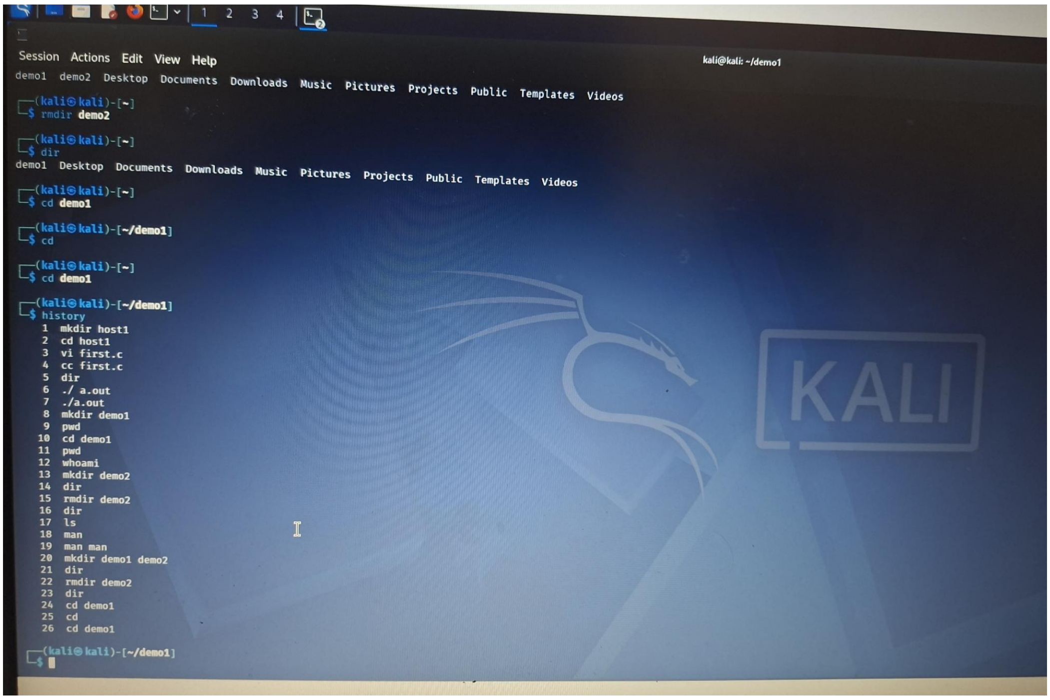
Sends output to a file and adds it at the end without deleting existing content.

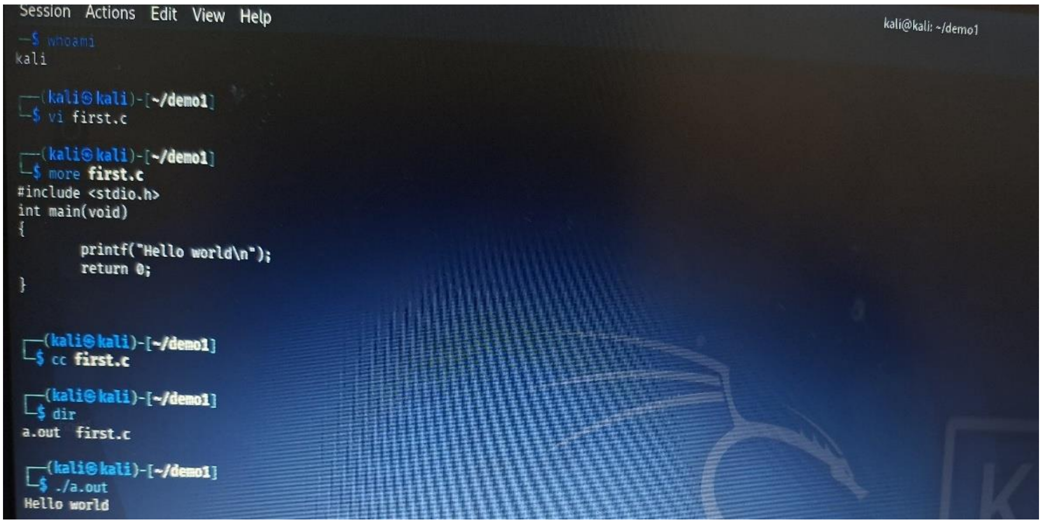
./program >> output.txt

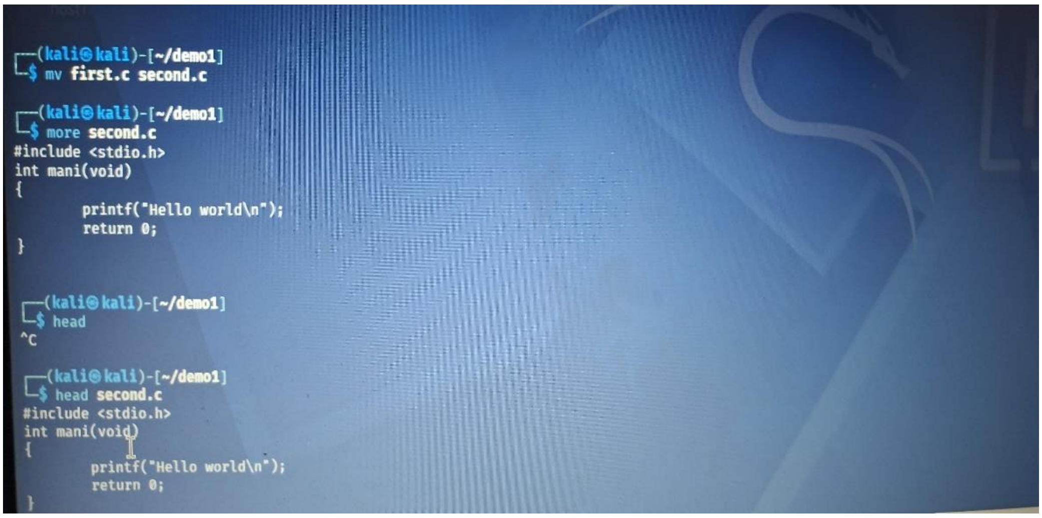
**<u>< Input redirection:</u>**

Takes input for the program from a file instead of the keyboard.

./program < input.txt


Session Actions Edit View Help
kali@kali: ~/demo1
demo1 demo2 Desktop Documents Downloads Music Pictures Projects Public Templates Videos
(kali㉿kali)-[~]
$ rmdir demo2
(kali㉿kali)-[~]
$ dir
demo1 Desktop Documents Downloads Music Pictures Projects Public Templates Videos
(kali㉿kali)-[~]
$ cd demo1
(kali㉿kali)-[~/demo1]
$ cd
(kali㉿kali)-[~]
$ cd demo1
(kali㉿kali)-[~/demo1]
$ history
1 mkdir host1
2 cd host1
3 vi first.c
4 cc first.c
5 dir
6 ./ a.out
7 ./a.out
8 mkdir demo1
9 pwd
10 cd demo1
11 pwd
12 whoami
13 mkdir demo2
14 dir
15 rmdir demo2
16 dir
17 ls
18 man
19 man man
20 mkdir demo1 demo2
21 dir
22 rmdir demo2
23 dir
24 cd demo1
25 cd
26 cd demo1
(kali㉿kali)-[~/demo1]
$
KALI


Session Actions Edit View Help
kali@kali: ~/demo1
$ whoami
kali
(kali㉿kali)-[~/demo1]
$ vi first.c
(kali㉿kali)-[~/demo1]
$ more first.c
#include <stdio.h>
int main(void)
{
printf("Hello world\n");
return 0;
}
(kali㉿kali)-[~/demo1]
$ cc first.c
(kali㉿kali)-[~/demo1]
$ dir
a.out first.c
(kali㉿kali)-[~/demo1]
$ ./a.out
Hello world




**Submitted By:** Reyansh Pandit
**SAP ID:** 590033602
**Batch:15**